TV Show ML Project

Objective:

To predict the popularity of a TV show based on its characteristics, using a large dataset of more than 160,000 records. The goal is to build a baseline model that can estimate the potential success of a show before it airs.

About the Data:

The dataset contains 29 columns (including the target variable 'popularity'). These include features such as the genres, air dates, production networks, and more.

Project Structure:

The project is divided into multiple Jupyter notebooks, each focusing on a distinct stage of the ML pipeline:

1. Data Preparation:

- Initial exploration of the data.
- Grouping large categorical values like networks and genres.
- Extracting relevant data from raw columns before grouping.
- Dropping irrelevant or high-NaN columns.
- Transforming datetime columns into year and month formats.

2. EDA (Exploratory Data Analysis):

- Analyzed distributions and patterns in the data.
- Found that some critical columns (e.g., vote_count, vote_average) had many zeros that should be interpreted as missing values.
- Identified temporal patterns where such zeros were most frequent.
- Used Spearman correlation due to data skewness.
- Applied log transformation to popularity and performed T-tests and ANOVA on categorical variables like type and genre to assess statistical differences.

3. Data Cleansing:

- Outlier detection performed.
- If outliers significantly affected distribution or correlation, they were replaced with NaN.
- Analyzed intersection between popularity outliers and other features.
- Retained outliers for features like number_of_seasons, number_of_episodes, and vote_count due to high overlap with popularity.
- Used KNN imputation to fill missing values.
- Kept two datasets to compare model performance post-imputation.

Feature Engineering:

- Created and encoded new features.
- Removed columns like show names that introduce high cardinality.

New Features:
- changed_name: 1 if the name differs from the original name.
- duplicate_name: 1 if the name appears multiple times.
- polarity, subjectivity, sentiment: Extracted using TextBlob.
- top_words: Indicator for top 15% most common words in overviews.
- overview_length: Word count of the show overview.
- *_num columns: Count of items in original multi-valued categorical fields (e.g., languages, networks).
- aired_time: Duration between first and last air date.
- avg_episodes_per_season: Self-explanatory.
- growth_index: TV access growth rate from 1985 onward (2.69% CAGR) including streaming platforms.

Result: Final dataset with 36 features.

Feature Selection:

- Used models with regularization or tree-based importance:
  - Lasso
  - Ridge
  - Gradient Boosting
  - Random Forest
- Used SelectFromModel to retain only features selected by all four models.
- Tuned alpha using Grid Search for Ridge and Lasso.
- Adjusted Lasso's alpha manually to retain at least 50% of features.

Model Selection & Fine-Tuning:

Evaluation Metrics:
- RMSLE (primary)
- MAE
- MSE
- RMSE

Models Tested:
- Linear Regression
- SVR (dropped due to high runtime and poor performance)
- Random Forest
- Gradient Boosting
- AdaBoost
- XGBoost

Methodology:
- Used cross-validation (CV) for all models.
- Ran all models with default parameters.
- Fine-tuned best-performing models with GridSearchCV.
- Chose RMSLE as the main metric due to the skewed target variable.

Top Models:
1. Random Forest (best RMSLE)
2. Gradient Boosting
3. XGBoost

Final Model Evaluation:

Evaluated best model (Random Forest) on a hold-out test set. Final predictions were generated and performance metrics were reported.

Summary:

The project provides a structured, end-to-end ML pipeline—from raw TV show data to final predictions. The analysis accounts for data quality, statistical validation, imputation techniques, and performance optimization. The final model offers a solid baseline for predicting a show's potential popularity using structured metadata.

Report:

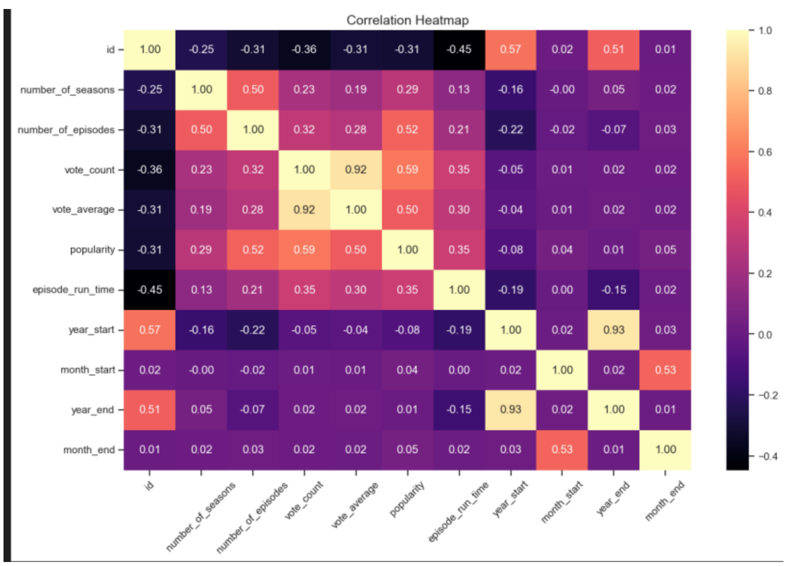
From the initial descriptive statistics, I observed that several columns contain a minimum value of 0. In particular, the **vote_count** and **vote_average** stood out, as a value of 0 in these fields likely indicates missing or unrecorded user engagement data rather than actual zeros. Recognizing their potential impact on downstream analysis, I flagged these two features for deeper examination. The remaining columns with zeros were addressed later during the data cleansing phase.

	id	number_of_seasons	number_of_episodes	vote_count	vote_average	popularity	episode_run_time	year_start	month_start	year_end	month_end
count	166254.000000	166254.000000	166254.000000	166254.00000	166254.000000	166254.000000	166254.000000	134657.000000	134657.000000	136508.000000	136508.000000
mean	110780.987447	1.554020	24.749191	13.48204	2.328815	5.967034	22.760571	2010.385401	6.267249	2011.559073	6.583475
std	76478.548602	2.960935	135.732600	192.16682	3.449142	42.317617	48.157980	13.924199	3.560971	13.487222	3.542973
min	1.000000	0.000000	0.000000	0.00000	0.000000	0.600000	0.000000	1917.000000	1.000000	1917.000000	1.000000
25%	45285.250000	1.000000	1.000000	0.00000	0.000000	0.600000	0.000000	2006.000000	3.000000	2007.000000	3.000000
50%	96968.000000	1.000000	6.000000	0.00000	0.000000	0.878000	0.000000	2015.000000	6.000000	2017.000000	6.000000
75%	196639.750000	1.000000	20.000000	1.00000	6.000000	2.487000	42.000000	2020.000000	9.000000	2021.000000	10.000000
max	251083.000000	240.000000	20839.000000	21857.00000	10.000000	3707.008000	6032.000000	2029.000000	12.000000	2024.000000	12.000000

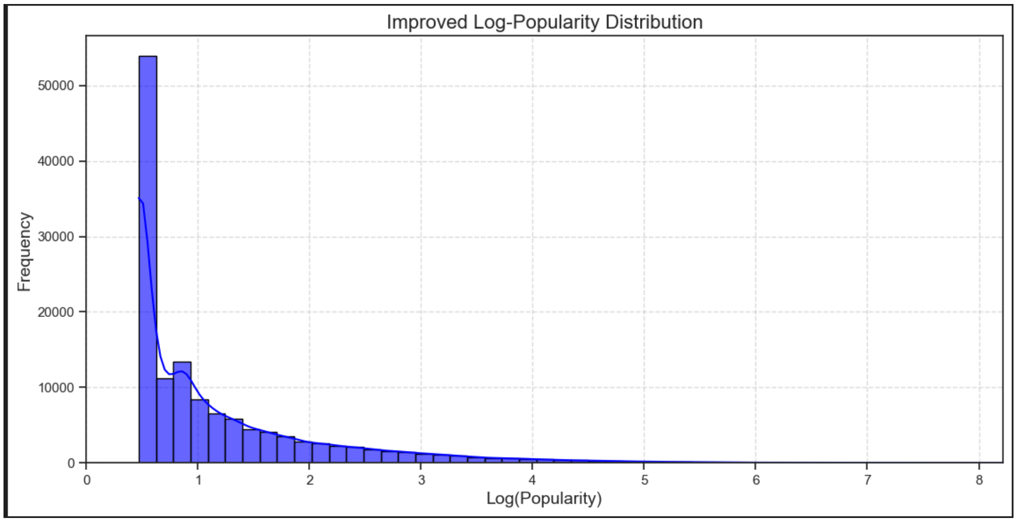
Correlation Heatmap

To explore relationships between numerical features, I generated a correlation heatmap. The analysis revealed several noteworthy patterns:

- **Vote count** and **popularity** show a strong positive correlation (~0.59), suggesting that higher popularity often coincides with greater user engagement.
- **Vote average** also correlates with **popularity**, but more moderately (~0.50), indicating that higher-rated shows tend to be more popular, though the effect is less pronounced.
- **Number of episodes** is positively correlated with **vote_count** and **popularity**, which could imply that longer-running shows have more exposure and thus attract more votes.



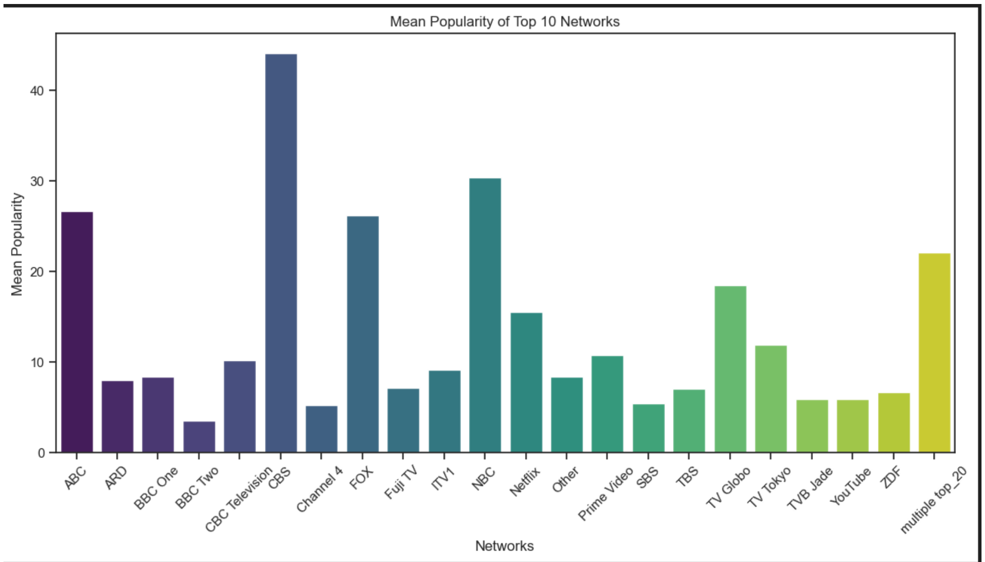
The popularity (target column) is very skewed I needed to use log transformation to capture it:



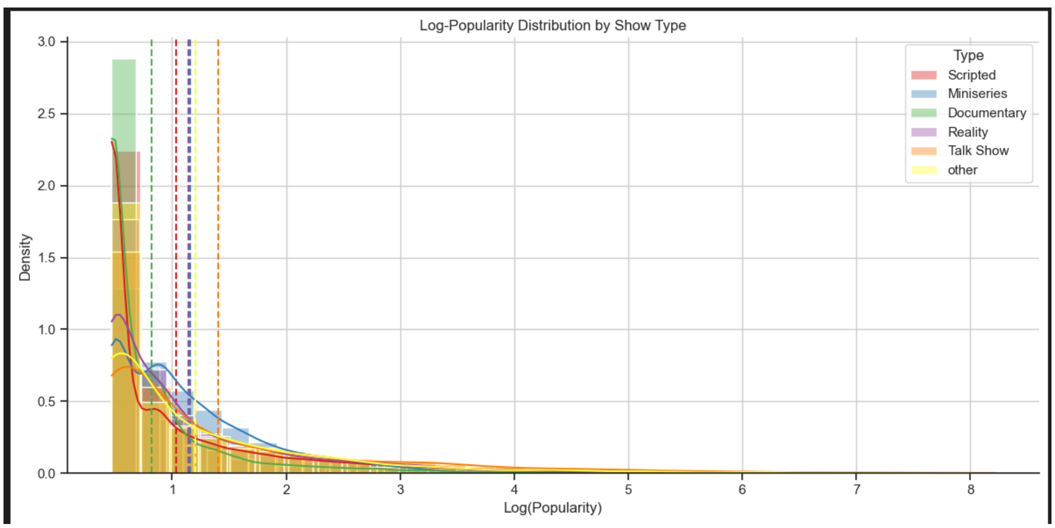
Skewness:

	skewness
number_of_episodes	55.262482
vote_count	47.838019
popularity	36.619365
episode_run_time	35.007164
number_of_seasons	14.033678
vote_average	0.965829
id	0.389139
month_end	-0.011423
month_start	-0.029918
year_start	-1.714173
year_end	-1.800749

The average popularity score for each of the top networks:



Anova for the type of show, the lines represent the group mean:



Data cleansing:

Outliers columns with count of total and their effect on the data:

	feature	outliers_cnt	distribution_changed	correlation_changed
0	id	0	-	-
1	number_of_seasons	54307	+	-
2	number_of_episodes	16710	+	+
3	vote_count	25957	+	+
4	vote_average	0	-	-
5	popularity	23880	+	-
6	episode_run_time	2852	+	-
7	year_start	9321	+	-
8	month_start	0	-	-
9	year_end	8938	+	-
10	month_end	0	-	-

<u>Popularity outlier:</u>

```
Total Popularity Outliers: 23880
Percentage of Popularity Outliers: 14.38%
```

The total present of columns outliers that are located in the popularity outliers' rows:

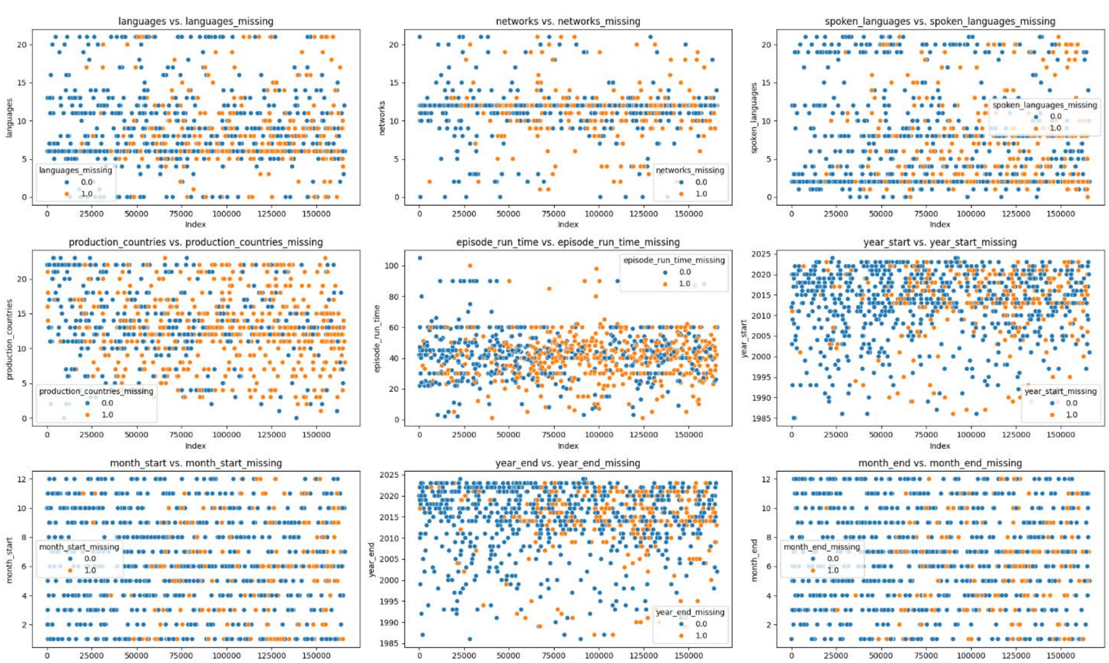
I decided to keep "number of seasons", "number of episodes"," vote count" outliers and change the rest to NAN.

	feature	sum_out	present_of_total
0	id	0	0.000000
1	number_of_seasons	10804	0.198943
2	number_of_episodes	9786	0.585637
3	vote_count	14341	0.552491
4	vote_average	0	0.000000
5	popularity	23880	1.000000
6	episode_run_time	545	0.191094
7	year_start	2256	0.242034
8	month_start	0	0.000000
9	year_end	1857	0.207765
10	month_end	0	0.000000

NAN values:

```
id : total missimg values : 0 by presentage: 0.0%
name : total missimg values : 5 by presentage: 0.003011068688498922%
number_of_seasons : total missimg values : 0 by presentage: 0.0%
number_of_episodes : total missimg values : 0 by presentage: 0.0%
original_language : total missimg values : 0 by presentage: 0.0%
vote_count : total missimg values : 0 by presentage: 0.0%
vote_average : total missimg values : 0 by presentage: 0.0%
overview : total missimg values : 74518 by presentage: 44.875763305912535%
adult : total missimg values : 0 by presentage: 0.0%
in_production : total missimg values : 0 by presentage: 0.0%
original_name : total missimg values : 5 by presentage: 0.003011068688498922%
popularity : total missimg values : 0 by presentage: 0.0%
type : total missimg values : 0 by presentage: 0.0%
status : total missimg values : 0 by presentage: 0.0%
genres : total missimg values : 67826 by presentage: 40.845748973225575%
languages : total missimg values : 57929 by presentage: 34.88563961121081%
networks : total missimg values : 68876 by presentage: 41.47807339781035%
origin_country : total missimg values : 0 by presentage: 0.0%
spoken_languages : total missimg values : 58690 by presentage: 35.34392426560034%
production_countries : total missimg values : 90029 by presentage: 54.21670059137389%
episode_run_time : total missimg values : 2852 by presentage: 1.7175135799197851%
year_start : total missimg values : 40951 by presentage: 24.661254772543874%
month_start : total missimg values : 31597 by presentage: 19.028147470100087%
year_end : total missimg values : 38715 by presentage: 23.314704855047154%
month_end : total missimg values : 29744 by presentage: 17.91224541414239%
```

NAN fill KNN: orange imply that there was missing data there



Feature engineering:

	Feature	Lasso	Ridge	GB	RF	Total
0	id	1	1	1	1	4
1	number_of_seasons	1	1	1	1	4
2	number_of_episodes	1	1	1	1	4
3	original_language	1	1	1	1	4
4	vote_count	1	1	1	1	4
5	vote_average	1	1	1	1	4
6	adult	0	1	0	1	2
7	in_production	0	1	1	1	3
8	type	0	1	1	1	3
9	status	0	1	0	1	2
10	genres	1	1	1	1	4
11	languages	0	1	1	1	3
12	networks	0	1	1	1	3
13	origin_country	1	1	1	1	4
14	spoken_languages	1	1	1	1	4
15	production_countries	1	1	1	1	4
16	episode_run_time	1	1	1	1	4
17	year_start	0	1	1	1	3
18	month_start	0	1	1	1	3
19	year_end	0	1	1	1	3
20	month_end	0	1	1	1	3
21	changed_name	0	1	0	1	2
22	name_is_duplicate	0	1	0	1	2
23	polarity	1	1	1	1	4
...						
31	networks_num	0	1	1	1	3
32	aired_time	1	1	1	1	4
33	avg_episodes_per_season	1	1	1	1	4
34	growth_index	1	1	1	1	4

The final columns I got after the feature selection

#	Column	Non-Null Count	Dtype
0	id	166054 non-null	int64
1	number_of_seasons	166054 non-null	Int64
2	number_of_episodes	166054 non-null	Int64
3	original_language	166054 non-null	int64
4	vote_count	166054 non-null	int64
5	vote_average	166054 non-null	float6
6	genres	166054 non-null	Int64
7	origin_country	166054 non-null	int64
8	spoken_languages	166054 non-null	Int64
9	production_countries	166054 non-null	Int64
10	episode_run_time	166054 non-null	Int64
11	polarity	166054 non-null	int64
12	subjectivity	166054 non-null	int64
13	overview_length	166054 non-null	int64
14	aired_time	166054 non-null	float6
15	avg_episodes_per_season	166054 non-null	Float6
16	growth_index	166054 non-null	float6

Model selection & fine tuning

The score of all the models : the Random forest have the best score of RMSLE

	model	MSE	RMSE	MAE	RMSLE
0	Linear Regression	1580.560500	39.756264	6.402439	0.969606
1	RandomForest	1280.304666	35.781345	4.605238	0.542753
2	ADABoost	16689.543785	129.188017	96.550460	3.464001
3	GBM	1264.498024	35.559781	4.910164	0.676927
4	XGBoost	1314.926794	36.261919	4.668367	0.586778
5	average_top3_models	1207.367802	34.747198	4.556477	0.577926

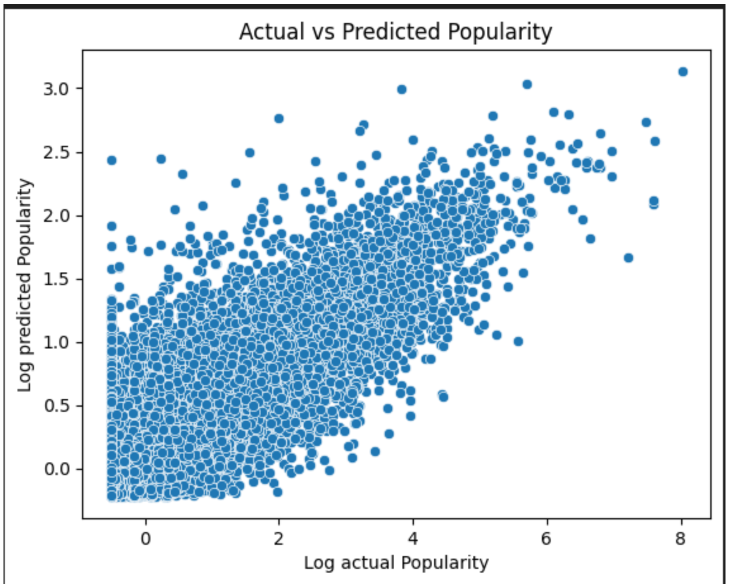
Fine tuning:

After the grid search I did for the Random forrest model hper parameters rhose are the best ones for the model:

	model	max_depth	max_features	min_samples_leaf	min_samples_split	n_estimators
0	RandomForest	30	None	4	8	80

Model evaluation:
I evaluated the model through a sample tets data set **:**

	Model	MSE	RMSE	MAE	R2	RMSLE
0	Random forest	1118.40749	33.4426	4.182637	0.37621	0.502673


Actual vs Predicted Popularity
Log predicted Popularity
Log actual Popularity
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
2
4
6
8